Andrea Arriola Gamboa

Bioinformatician | Genomics, Transcriptomics & Data Analysis

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PROFILE

Bioinformatician with an MSc in Bioinformatics from the University of Bologna (107/110) and hands-on experience across genomics and transcriptomics. I have built NGS pipelines from raw whole-genome sequencing data to VCF files on a computing cluster, and analysed bulk and single-cell RNA-seq data, including a benchmark of differential-expression methods validated against bulk RNA-seq with precision and recall. I work in R, Python and Bash, use Git, Conda and Docker for reproducible analyses, and apply statistics and machine learning with careful evaluation (cross-validation, ROC/AUC).

PROFESSIONAL EXPERIENCE

Bioinformatics Intern | *University of Padova (Bicciato Lab)*
Feb 2026 – Sept 2026

- Built in R a single-cell RNA-seq analysis pipeline (preprocessing, quality control, normalization, clustering, differential expression) to characterize keratinocytes from EEC syndrome, a rare disease, and benchmarked differential-expression methods (Seurat tests incl. MAST, NEBULA-LN, muscat, pseudobulk DESeq2/edgeR) against bulk RNA-seq as a surrogate ground truth, using precision and recall.

Research Intern | *National Institute of Genomic Medicine in Mexico (INMEGEN)*
Jul 2023 – Apr 2024

- Developed and ran NGS pipelines in Bash on the institute's computing cluster, processing raw whole-genome sequencing data through quality control, alignment and variant calling to VCF files, for a population-genomics project on Indigenous populations in Latin America.
- Data-curation contribution acknowledged in Miron-Toruno et al. (2025), "Genome-Wide Selection Scans in Mexican Indigenous Populations Reveal Recent Signatures of Pathogen and Diet Adaptation," Genome Biology and Evolution 17(3), DOI: 10.1093/gbe/evaf043.

Bioinformatics Analyst | *Core Lab: Genomics*
Aug 2022 – Nov 2022

- Performed statistical analysis of bulk RNA-seq gene-expression data in R, including differential-expression testing, to characterize stress responses in *Aspergillus niger* and support metabolite-production research.

EDUCATION

MSc. Bioinformatics (Class LM-6 – Biology)
Alma Mater Studiorum – Università di Bologna, Italy | Oct 2024 – Sept 2026
Degree completed September 2026, final grade 107/110. Thesis: "Single-Cell Transcriptomic Characterization of EEC Syndrome Keratinocytes and Benchmarking of Differential Expression Methods" (Bicciato Lab, University of Padova).

BSc. Biotechnology Engineering
Tecnológico de Monterrey, Mexico | Aug 2019 – Jun 2023
Functional Genomics and Synthetic Biology Specialization

PROJECTS

Signal Peptide Detection – Machine Learning Pipeline | *Laboratory of Bioinformatics 2 course, 5-person team project*

- Collaborated in a 5-person team, using Git/GitHub for version control, to build and compare a von Heijne statistical model and an SVM classifier for signal peptide prediction, validating results via 5-fold cross-validation with ROC/AUC, sensitivity and specificity; the SVM model reached 97% accuracy.

Population-Genomics Pipeline Design – Pooled WGS | *Applied Genomics course (grade 29/30)*

- Designed a pooled-WGS pipeline (QC, alignment, variant calling, windowed FST) for honeybee adaptation.

Diagnostic Classification – Breast Cancer (Python) | *Applied Machine Learning course*

- Malignant-vs-benign tumor classifier (Breast Cancer Wisconsin); ROC/AUC, sensitivity, specificity.

SKILLS

- R/Bioconductor (DESeq2, edgeR, Seurat, MAST, muscat), Python (pandas, ML) & Bash/Linux
- NGS processing to VCF on a computing cluster; bulk & single-cell RNA-seq analysis
- Git/GitHub, Docker (incl. Dockerfiles), Conda, R Markdown; SQL, Redis & REST APIs (UniProt, NCBI, PDB)
- Statistics & machine learning: hypothesis testing, cross-validation, ROC/AUC, precision/recall

LANGUAGES

Spanish — Native
English — Proficient (C1)
Italian — Upper-Intermediate (B2)

I authorize the processing of my personal data contained in this CV for the purpose of recruitment in accordance with Regulation (EU) 2016/679 (GDPR).